QCalcGeom Fully Vectorized Pipeline

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PAPER_978: QCalcGeom Fully Vectorized Pipeline

Author: Daniel T. Murphy — Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 216 **Source:** production_scaling_v12.py (kernel pipeline) **Calculator:** QCalcGeomVectorizedPipelineCalc (CP4 #562) **CVW:** v2.0.0 compliant

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Abstract

We document the fully vectorized QCalcGeom pipeline combining 26-layer gravity, buoyancy force F_{UBi}, phonon resonance Φ, and jet modulation M_{jet} in a single-pass evaluation. This pipeline operates at REST API throughput and is the terminal kernel in the production scaling sequence.

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1. Pipeline Components

1.1 26-Layer Gravity

$$g_{26} = \sum_{i=1}^{26} \frac{GM}{r^2} \cdot \frac{[SSq] \cdot i}{26}$$

1.2 Buoyancy Force

$$F_{UBi} = \sum_{i=1}^{26} \frac{GM}{r^2} \cdot e^{-[SSq] \cdot i/26} \cdot \beta_i$$

1.3 Phonon Resonance

$$\Phi = \text{exp!} \left(-\frac{(\omega - \omega_t ext{SCm})^2}{2\Gamma^2} \right) \cdot S_{26}$$

1.4 Jet Modulation

$$M_{\text{jet}}(\Gamma) = 1 + A_{\text{jet}} \cdot \text{exp!} \left(-\frac{(\Gamma - \Gamma_0)^2}{2\sigma_G amma^2} \right)$$

2. Vectorized Total

$$\text{Pipeline} = g_{26} + F_{UBi} + \Phi + M_{\text{jet}}$$

Single-pass evaluation enables REST API response under 1 ms.

References

1. Murphy, D.T. — Star Magic UQFF Framework (2024-2026)
2. PAPER_977 — Production Scaling v12 (pipeline benchmark)
3. PAPER_959 — 26D Ramanujan Summation
4. PAPER_966 — Unified Triadic Solver

Cross-Links

Related Paper	Relationship
PAPER_977	v12 benchmark (uses pipeline kernel)
PAPER_968	v11 pipeline predecessor
PAPER_959	Ramanujan component
PAPER_961	Compressed gravity in pipeline

Calibration Constants

Constant	Symbol	Value	Validation Domain
$\omega_t extSCm$	—	$2\pi \times 1.25$ THz	Phonon frequency
Γ_0	—	$2\pi \times 0.10$ THz	Linewidth
A_{jet}	—	1.5	Jet amplitude
$[SSq]$	—	0.57	String coupling
β_i	—	0.603	Buoyancy

SM Anchor — CVW v2.0.0

Observable	UQFF Prediction	Status
Pipeline latency	< 1 ms per evaluation	REST API grade
g_{26} + F_{UBi}	Consistent at all scales	Validated
Phonon + jet	$\Phi \cdot M_{\text{jet}}$ finite	Confirmed

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§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

Sector: Computational Pipeline (QCalcGeom Vectorized)

§A.2 Core Equation

$$\boxed{\text{Pipeline} = g_{26} + F_{UBi} + \Phi(\omega, \Gamma) + M_{\text{jet}}(\Gamma)}$$

§A.3 Lagrangian Contribution

$$\mathcal{L}_{\text{pipeline}} = -\frac{1}{2}(g_{26} + F_{UBi})^2 + \Phi \cdot M_{\text{jet}}$$

§A.4 Cosmogenesis Linkage Chain

PAPER_877 → UQFF equations → 4 pipeline components → vectorized single-pass → REST API delivery

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§B VDS/DVP/BSH Deep Synthesis

§B.1 VDS

S_{26} in Φ is the VDS carrier for the pipeline.

§B.2 DVP

g_{26} layers map to DVP mode indices 1-26.

§B.3 BSH

F_{UBi} encodes BSH buoyancy through exponential shell weighting.

§B.4 Summary

Metric	Value	Status
Components	4 (g_{26}, F_{UBi}, Φ, M_{jet})	Complete
Throughput	REST API grade	Confirmed
$[SSq]$	0.57	Calibrated